

GNSS spoofing detection

Signature tests on receiver observables

Received power and
automatic gain control

Correlation function
distortion

Code and carrier
consistency

Cross satellite
residuals

Antenna array
spatial processing

Message
authentication

Detectors inferred from labelled recordings

Single epoch
classifiers

Sequence models
over a window

Authentic class
only models

Generative
augmentation

Physics constrained
training

Guarantees proved in advance

Randomized smoothing,
isotropic

Anisotropic
certification

Position domain
interpretation

*names one physical signature;
a spoofer that respects it passes*

*needs labelled attacks;
describes the attacks recorded*

*needs a noise model;
silent outside its own radius*